

Finite Elements for Differential Geometry - Exercise 3

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Exercise 9:

Investigate the linear Kirchhoff–Love shell example `Linear_Koiter_shell_HHJ.ipynb` for the given thicknesses. What do you observe when interpolating the membrane strains into the Regge space (`HCurlCurl(mesh,order=order-1)`)?

Exercise 10:

Prove Lemma 5.2.1 showing the split of the Cauchy–Green strain tensor of the deformed and initial configuration by inserting the ansatz (5.2.1) into the definition of the Cauchy–Green strain tensor and comparing with the objects in (5.1.5).

Exercise 11:

Prove Lemma 5.3.1 that the three-field formulation (5.3.2) is equivalent to the two-field formulation (5.3.3) by computing the critical points of the Lagrangians and eliminating κ^{diff} .

Hint: Don't try to compute the variation with respect to u in detail.